Control Systems Theory

Mathematical Foundations for the

Triple Inverted Pendulum on a Cart

State-Space Modeling · *Lagrangian Dynamics*

PID Control · *LQR Optimal Control* · *MPC*

Extended Kalman Filter · *Unscented Kalman Filter*

MATLAB / Simulink Implementation Reference

Prepared as a technical reference for the portfolio project
`github.com/aapaul29/inverted-pendulum-control`

Contents

1 System Description and Notation

1.1 Physical Setup

The system consists of a cart of mass M free to slide along a horizontal rail, with three rigid pendulum links attached in series. Link i has mass m_i and length ℓ_i, for $i = 1, 2, 3$. The cart is actuated by a horizontal force u.

State Vector (interleaved convention):

$$\boldsymbol{x} = \begin{bmatrix} x \\ \dot{x} \\ \theta_1 \\ \dot{\theta}_1 \\ \theta_2 \\ \dot{\theta}_2 \\ \theta_3 \\ \dot{\theta}_3 \end{bmatrix} \in \mathbb{R}^8$$

where x is cart displacement and θ_i is the absolute angle of link i from vertical.

1.2 Coordinate Convention

All angles θ_i are measured from the upright vertical. The equilibrium of interest is:

$$\boldsymbol{x}^* = \mathbf{0}_{8\times 1}, \qquad u^* = 0.$$

This equilibrium is *open-loop unstable*: any infinitesimal perturbation causes the links to fall.

2 Lagrangian Dynamics and Equations of Motion

2.1 Lagrangian Formulation

The equations of motion are derived via the Euler–Lagrange equations:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_k} - \frac{\partial \mathcal{L}}{\partial q_k} = Q_k,$$

where $\mathcal{L} = T - V$ is the Lagrangian (kinetic minus potential energy), $\boldsymbol{q} = [x,\, \theta_1,\, \theta_2,\, \theta_3]^\top$ are the generalised coordinates, and Q_k are generalised forces.

2.1.1 Kinetic Energy

The position of the centre of mass of link i (length ℓ_i, assumed uniform) is:

$$x_{c,i} = x + \sum_{j=1}^{i-1} \ell_j \sin\theta_j + \tfrac{\ell_i}{2}\sin\theta_i, \quad y_{c,i} = \sum_{j=1}^{i-1} \ell_j \cos\theta_j + \tfrac{\ell_i}{2}\cos\theta_i.$$

The total kinetic energy is:

$$T = \tfrac{1}{2}M\dot{x}^2 + \sum_{i=1}^{3} \tfrac{1}{2}m_i\left(\dot{x}_{c,i}^2 + \dot{y}_{c,i}^2\right) + \sum_{i=1}^{3} \tfrac{1}{2}I_i\dot{\theta}_i^2,$$

where $I_i = \frac{1}{12}m_i\ell_i^2$ is the moment of inertia about the link's centre of mass.

2.1.2 Potential Energy

$$V = \sum_{i=1}^{3} m_i g\, y_{c,i}.$$

2.1.3 Compact Matrix Form

The equations of motion take the standard manipulator form:

$$\boldsymbol{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) = \boldsymbol{B}\,u,$$

where:

- $\boldsymbol{M}(\boldsymbol{q}) \in \mathbb{R}^{4\times 4}$ is the symmetric positive-definite inertia matrix,
- $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}}) \in \mathbb{R}^{4\times 4}$ contains Coriolis and centrifugal terms,
- $\boldsymbol{G}(\boldsymbol{q}) \in \mathbb{R}^{4}$ is the gravity vector,
- $\boldsymbol{B} = [1,\ 0,\ 0,\ 0]^\top$ maps the scalar force u to generalised coordinates.

3 Linearisation and State-Space Representation

3.1 Nonlinear State-Space Form

Defining $\boldsymbol{x} \in \mathbb{R}^8$ as above and $\boldsymbol{u} = u \in \mathbb{R}$, the dynamics are written as:

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, u), \qquad \boldsymbol{y} = h(\boldsymbol{x}),$$

where $f : \mathbb{R}^8 \times \mathbb{R} \to \mathbb{R}^8$ encodes the full nonlinear ODEs.

3.2 First-Order Taylor Linearisation

Expanding f around the equilibrium $(\boldsymbol{x}^*, u^*) = (0, 0)$:

$$\dot{\boldsymbol{x}} \approx \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}u,$$

$$\boldsymbol{A} = \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}^*,\,u^*} \in \mathbb{R}^{8\times 8}, \qquad \boldsymbol{B} = \left.\frac{\partial f}{\partial u}\right|_{\boldsymbol{x}^*,\,u^*} \in \mathbb{R}^{8\times 1}.$$

At the upright equilibrium, $\cos\theta_i \approx 1$ and $\sin\theta_i \approx \theta_i$, so the inertia matrix reduces to a constant and all Coriolis terms vanish.

3.3 Output Equation

If only the cart position and three angles are measured:

$$\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x}, \qquad \boldsymbol{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \in \mathbb{R}^{4\times 8}.$$

3.4 Controllability and Observability

Definition 3.1. The system $(\boldsymbol{A}, \boldsymbol{B})$ is *controllable* if and only if the controllability matrix

$$\mathcal{C} = \begin{bmatrix} \boldsymbol{B} & \boldsymbol{A}\boldsymbol{B} & \boldsymbol{A}^2\boldsymbol{B} & \cdots & \boldsymbol{A}^7\boldsymbol{B} \end{bmatrix} \in \mathbb{R}^{8\times 8}$$

has rank 8.

Definition 3.2. The system $(\boldsymbol{C}, \boldsymbol{A})$ is *observable* if and only if the observability matrix

$$\mathcal{O} = \begin{bmatrix} \boldsymbol{C}^\top & \boldsymbol{A}^\top \boldsymbol{C}^\top & \cdots & (\boldsymbol{A}^\top)^7 \boldsymbol{C}^\top \end{bmatrix}^\top \in \mathbb{R}^{32\times 8}$$

has rank 8.

Both conditions *must* hold before any model-based controller or observer can be designed. In MATLAB: `rank(ctrb(A,B))` and `rank(obsv(A,C))`.

4 PID Control

4.1 Classical PID Structure

A PID controller generates a control signal from the error $e(t) = r(t) - y(t)$:

$$u(t) = K_p\, e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d\, \frac{de(t)}{dt},$$

with transfer function:

$$C(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s}.$$

4.2 Application to the Triple Pendulum

For the triple pendulum, a PID loop is applied per-link using the angle error $e_i = -\theta_i$:

$$u_i(t) = K_{p,i}\theta_i + K_{d,i}\dot{\theta}_i + K_{i,i}\int_0^t \theta_i\, d\tau.$$

The total force command is the superposition:

$$u(t) = \sum_{i=1}^{3} u_i(t).$$

4.3 Limitations

Why PID is insufficient here:

1. Ignores coupling between links: each loop treats other states as disturbances.
2. Cart position x is not naturally regulated by angle-only feedback.
3. Tuning 9+ parameters for 3 interacting loops is non-systematic.
4. No formal optimality guarantee.

LQR addresses all four limitations through full-state feedback.

4.4 Tuning Methods

4.4.1 Ziegler–Nichols (Frequency Response)

1. Set $K_i = K_d = 0$. Increase K_p until sustained oscillation occurs at gain K_u and period T_u.
2. Apply: $K_p = 0.6K_u$, $K_i = 1.2K_u/T_u$, $K_d = 0.075K_uT_u$.

4.4.2 Root Locus / Pole Placement

Design $C(s)$ so that closed-loop poles lie in a desired region of the left half-plane. For a second-order target with damping ζ and natural frequency ω_n, desired poles are $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$.

5 LQR Optimal State-Feedback Control

5.1 Problem Formulation

Given the linear system $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}u$, find the state-feedback gain $\boldsymbol{K} \in \mathbb{R}^{1\times 8}$ such that $u = -\boldsymbol{K}\boldsymbol{x}$ minimises the infinite-horizon quadratic cost:

$$J = \int_0^\infty \left(\boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x} + u^\top \boldsymbol{R}\, u\right) dt,$$

where:

- $\boldsymbol{Q} = \boldsymbol{Q}^\top \succeq 0 \in \mathbb{R}^{8\times 8}$ penalises state deviation,
- $\boldsymbol{R} = \boldsymbol{R}^\top \succ 0 \in \mathbb{R}^{1\times 1}$ penalises control effort.

5.2 Algebraic Riccati Equation (ARE)

Theorem 5.1 (LQR Optimality)**.** If $(\boldsymbol{A}, \boldsymbol{B})$ is stabilisable and $(\boldsymbol{A}, \boldsymbol{Q}^{1/2})$ is detectable, there exists a unique symmetric positive-semidefinite solution $\boldsymbol{P}$ to the ARE:

$$\boldsymbol{A}^\top \boldsymbol{P} + \boldsymbol{P}\boldsymbol{A} - \boldsymbol{P}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top \boldsymbol{P} + \boldsymbol{Q} = \boldsymbol{0}.$$

The optimal gain is then:

$$\boldsymbol{K}^* = \boldsymbol{R}^{-1}\boldsymbol{B}^\top \boldsymbol{P}.$$

The closed-loop system $\dot{\boldsymbol{x}} = (\boldsymbol{A} - \boldsymbol{B}\boldsymbol{K}^*)\boldsymbol{x}$ is asymptotically stable.

5.3 Weighting Matrix Selection

A standard starting point is *Bryson's rule*:

$$Q_{ii} = \frac{1}{x_{i,\max}^2}, \qquad R = \frac{1}{u_{\max}^2},$$

where $x_{i,\max}$ and $u_{\max}$ are the maximum acceptable values for state i and the control input.

Tuning intuition:

- Increase Q_{ii} to penalise excursions in state i more heavily.
- Increase R to reduce control effort (slower response, less actuator wear).
- The ratio $\boldsymbol{Q}/R$ governs the speed-effort tradeoff.

5.4 LQR vs. Pole Placement

For an 8-state system, placing all 8 closed-loop poles manually is impractical: small changes in desired pole locations can yield wildly different gain values. LQR provides a *systematic, physically interpretable alternative* via the $\boldsymbol{Q}$ and $\boldsymbol{R}$ matrices.

5.5 Guaranteed Robustness Margins

For a SISO LQR design, the optimal gain has guaranteed phase margin $\geq 60°$ and gain margin $[\frac{1}{2}, \infty)$. This is a fundamental property of LQR that PID does not inherit automatically.

6 Model Predictive Control (MPC)

6.1 Concept

MPC solves a finite-horizon optimal control problem *at every time step*, applies only the first control action, then re-optimises at the next step (receding horizon). This allows:

- Explicit handling of state and input constraints.
- Preview of future reference trajectories.
- Online adaptation to model changes.

6.2 Finite-Horizon Problem

At time t, given current state $\hat{\boldsymbol{x}}_t$, solve:

$$\min_{\{u_k\}_{k=0}^{N-1}} \sum_{k=0}^{N-1} \left(\boldsymbol{x}_k^\top \boldsymbol{Q} \boldsymbol{x}_k + u_k^\top \boldsymbol{R}\, u_k \right) + \boldsymbol{x}_N^\top \boldsymbol{P} \boldsymbol{x}_N$$

subject to:

$$\boldsymbol{x}_{k+1} = \boldsymbol{A}_d \boldsymbol{x}_k + \boldsymbol{B}_d u_k, \quad \boldsymbol{x}_k \in \mathcal{X}, \quad u_k \in \mathcal{U},$$

where $\boldsymbol{A}_d, \boldsymbol{B}_d$ are the discrete-time system matrices (via zero-order hold with sample time T_s), N is the prediction horizon, $\boldsymbol{P}$ is the LQR terminal cost, and $\mathcal{X}$,$\mathcal{U}$ are polyhedral constraint sets.

6.3 Discrete-Time Model

Discretise the continuous system with sample period T_s using the matrix exponential:

$$\boldsymbol{A}_d = e^{\boldsymbol{A}T_s}, \qquad \boldsymbol{B}_d = \int_0^{T_s} e^{\boldsymbol{A}\tau} \boldsymbol{B}\, d\tau \approx \left(\boldsymbol{A}^{-1} \left(e^{\boldsymbol{A}T_s} - \boldsymbol{I} \right) \right) \boldsymbol{B}.$$

6.4 QP Reformulation

Stacking predictions over the horizon yields the condensed QP:

$$\min_{\boldsymbol{U}} \tfrac{1}{2} \boldsymbol{U}^\top \boldsymbol{H} \boldsymbol{U} + \boldsymbol{f}^\top \boldsymbol{U} \quad \text{s.t.} \quad \boldsymbol{G}\boldsymbol{U} \le \boldsymbol{w} + \boldsymbol{S} \hat{\boldsymbol{x}}_t,$$

where $\boldsymbol{U} = [u_0, \ldots, u_{N-1}]^\top \in \mathbb{R}^N$, and $\boldsymbol{H}, \boldsymbol{f}, \boldsymbol{G}, \boldsymbol{S}$ are derived from $\boldsymbol{Q}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{A}_d, \boldsymbol{B}_d$. This QP is solved at every sample with a dedicated solver (e.g. OSQP).

6.5 MPC vs. LQR

Property	**LQR**	**MPC**
State constraints	No	Yes
Input constraints	No	Yes
Computational cost	Low (offline)	High (online QP)
Horizon	Infinite	Finite N
Reference preview	No	Yes
Optimality	Global (unconstrained)	Local (constrained)

7 State Estimation: Extended Kalman Filter

7.1 Motivation

In practice, only a subset of states is measurable and measurements are corrupted by noise:

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, u) + \boldsymbol{L}\,\boldsymbol{w}, \quad \boldsymbol{y} = h(\boldsymbol{x}) + \boldsymbol{v},$$

where $\boldsymbol{w} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}_w)$ is process noise and $\boldsymbol{v} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_v)$ is measurement noise.

7.2 EKF Algorithm

The EKF handles nonlinear f and h by linearising at each step.

7.2.1 Prediction Step

$$\hat{\boldsymbol{x}}_{k|k-1} = f(\hat{\boldsymbol{x}}_{k-1|k-1}, u_{k-1}), \tag{1}$$
$$\boldsymbol{P}_{k|k-1} = \boldsymbol{F}_{k-1}\boldsymbol{P}_{k-1|k-1}\boldsymbol{F}_{k-1}^\top + \boldsymbol{L}\boldsymbol{Q}_w\boldsymbol{L}^\top, \tag{2}$$

where $\boldsymbol{F}_{k-1} = \left.\dfrac{\partial f}{\partial \boldsymbol{x}}\right|_{\hat{\boldsymbol{x}}_{k-1|k-1}}$ is the Jacobian.

7.2.2 Update Step

$$\boldsymbol{S}_k = \boldsymbol{H}_k\boldsymbol{P}_{k|k-1}\boldsymbol{H}_k^\top + \boldsymbol{R}_v, \tag{3}$$
$$\boldsymbol{K}_k = \boldsymbol{P}_{k|k-1}\boldsymbol{H}_k^\top\boldsymbol{S}_k^{-1}, \tag{4}$$
$$\hat{\boldsymbol{x}}_{k|k} = \hat{\boldsymbol{x}}_{k|k-1} + \boldsymbol{K}_k\big(\boldsymbol{y}_k - h(\hat{\boldsymbol{x}}_{k|k-1})\big), \tag{5}$$
$$\boldsymbol{P}_{k|k} = (\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H}_k)\boldsymbol{P}_{k|k-1}, \tag{6}$$

where $\boldsymbol{H}_k = \left.\dfrac{\partial h}{\partial \boldsymbol{x}}\right|_{\hat{\boldsymbol{x}}_{k|k-1}}$.

7.3 Limitations of the EKF

- Jacobian linearisation discards higher-order terms; accuracy degrades for highly nonlinear systems.
- May diverge if initial estimate or covariance is far from the truth.
- Requires analytical Jacobians (or numerical differentiation).
- The propagated covariance can become non-positive-definite in poorly conditioned cases.

8 State Estimation: Unscented Kalman Filter

8.1 Unscented Transform

The UKF replaces Jacobian linearisation with the *unscented transform*: a set of $2n+1$ deterministically chosen *sigma points* that capture the mean and covariance of a Gaussian distribution exactly to the *third order* of a Taylor expansion. For $n = 8$ states:

8.1.1 Sigma Points

Given mean $\hat{\boldsymbol{x}}$ and covariance $\boldsymbol{P}$, compute the matrix square root $\boldsymbol{S} = \mathrm{chol}((n+\lambda)\boldsymbol{P})$:

$$\boldsymbol{\chi}_0 = \hat{\boldsymbol{x}},$$

$$\boldsymbol{\chi}_i = \hat{\boldsymbol{x}} + \boldsymbol{S}_i, \quad i = 1, \ldots, n,$$
$$\boldsymbol{\chi}_{n+i} = \hat{\boldsymbol{x}} - \boldsymbol{S}_i, \quad i = 1, \ldots, n,$$

where $\boldsymbol{S}_i$ denotes the i-th column of $\boldsymbol{S}$ and $\lambda = \alpha^2(n+\kappa) - n$.

8.1.2 Weights

$$W_0^{(m)} = \frac{\lambda}{n+\lambda}, \quad W_0^{(c)} = \frac{\lambda}{n+\lambda} + (1 - \alpha^2 + \beta),$$
$$W_i^{(m)} = W_i^{(c)} = \frac{1}{2(n+\lambda)}, \quad i = 1, \ldots, 2n.$$

Typical values: $\alpha = 10^{-3}$, $\beta = 2$, $\kappa = 0$.

8.2 UKF Algorithm

8.2.1 Prediction Step

1. Generate $2n+1$ sigma points from $\hat{\boldsymbol{x}}_{k-1|k-1}$, $\boldsymbol{P}_{k-1|k-1}$.
2. Propagate each sigma point through the nonlinear dynamics: $\boldsymbol{\chi}_i^- = f(\boldsymbol{\chi}_i, u_{k-1})$.
3. Reconstruct predicted mean and covariance:

$$\hat{\boldsymbol{x}}_{k|k-1} = \sum_{i=0}^{2n} W_i^{(m)} \boldsymbol{\chi}_i^-,$$

$$\boldsymbol{P}_{k|k-1} = \sum_{i=0}^{2n} W_i^{(c)} \big(\boldsymbol{\chi}_i^- - \hat{\boldsymbol{x}}_{k|k-1}\big)\big(\boldsymbol{\chi}_i^- - \hat{\boldsymbol{x}}_{k|k-1}\big)^\top + \boldsymbol{Q}_w.$$

8.2.2 Update Step

1. Generate new sigma points from $\hat{\boldsymbol{x}}_{k|k-1}$, $\boldsymbol{P}_{k|k-1}$.
2. Propagate through measurement function: $\boldsymbol{\gamma}_i = h(\boldsymbol{\chi}_i^-)$.
3. Compute predicted measurement mean: $\hat{\boldsymbol{y}}_k = \sum_{i=0}^{2n} W_i^{(m)} \boldsymbol{\gamma}_i$.
4. Compute innovation covariance and cross-covariance:

$$\boldsymbol{S}_k = \sum_{i=0}^{2n} W_i^{(c)} (\boldsymbol{\gamma}_i - \hat{\boldsymbol{y}}_k)(\boldsymbol{\gamma}_i - \hat{\boldsymbol{y}}_k)^\top + \boldsymbol{R}_v,$$
$$\boldsymbol{T}_k = \sum_{i=0}^{2n} W_i^{(c)} (\boldsymbol{\chi}_i^- - \hat{\boldsymbol{x}}_{k|k-1})(\boldsymbol{\gamma}_i - \hat{\boldsymbol{y}}_k)^\top.$$

5. Kalman gain and update:

$$\boldsymbol{K}_k = \boldsymbol{T}_k \boldsymbol{S}_k^{-1}, \quad \hat{\boldsymbol{x}}_{k|k} = \hat{\boldsymbol{x}}_{k|k-1} + \boldsymbol{K}_k(\boldsymbol{y}_k - \hat{\boldsymbol{y}}_k), \quad \boldsymbol{P}_{k|k} = \boldsymbol{P}_{k|k-1} - \boldsymbol{K}_k \boldsymbol{S}_k \boldsymbol{K}_k^\top.$$

8.3 UKF vs. EKF: Summary

Property	EKF	UKF
Nonlinearity handling	First-order linearisation	Third-order (sigma points)
Jacobians required	Yes	No
Accuracy	Lower for strong nonlinearities	Higher
Computational cost	$\mathcal{O}(n^3)$ (Cholesky)	$\mathcal{O}(n^3)$ (comparable)
Implementation complexity	Medium	Medium–High
Divergence risk	Higher	Lower

Why UKF for the triple pendulum: The nonlinear dynamics involve products of $\sin\theta_i$, $\cos\theta_i$, and $\dot\theta_i^2$. For large angles or fast transients (especially open-loop), the first-order approximation in the EKF introduces significant error. The UKF's sigma-point approach propagates the distribution more accurately without requiring Jacobians.

9 LQG: Combining LQR and the Kalman Filter

9.1 Separation Principle

When the system is linear and noise is Gaussian, the optimal control and estimation problems *decouple*:

- Design the LQR gain $\boldsymbol{K}$ ignoring noise.
- Design the Kalman filter gain $\boldsymbol{L}$ ignoring control.
- Combine: $u = -\boldsymbol{K}\hat{\boldsymbol{x}}$.

The resulting LQG controller is optimal. For nonlinear systems (triple pendulum), the separation principle is applied approximately by replacing the Kalman filter with an EKF or UKF.

9.2 Closed-Loop Architecture

$$\begin{array}{ccccc} r & \longrightarrow & \boxed{\text{LQR}: -\boldsymbol{K}} & \xrightarrow{u} & \boxed{\text{Plant}: f(\boldsymbol{x},u)} \\ & & \uparrow & & \downarrow \boldsymbol{y}_{\text{noisy}} \\ & & \hat{\boldsymbol{x}} & \longleftarrow & \boxed{\text{UKF}} \end{array}$$

10 Stability Analysis

10.1 Lyapunov Stability

Theorem 10.1 (Lyapunov's Second Method)**.** The equilibrium $\boldsymbol{x} = \boldsymbol{0}$ of $\dot{\boldsymbol{x}} = f(\boldsymbol{x})$ is asymptotically stable if there exists $V : \mathbb{R}^n \to \mathbb{R}$ with:

$$V(\boldsymbol{0}) = 0, \quad V(\boldsymbol{x}) > 0\ \forall\, \boldsymbol{x} \neq \boldsymbol{0}, \quad \dot V(\boldsymbol{x}) = \nabla V \cdot f(\boldsymbol{x}) < 0\ \forall\, \boldsymbol{x} \neq \boldsymbol{0}.$$

For the LQR-controlled linear system, the quadratic function $V(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{P}\boldsymbol{x}$ serves as a Lyapunov function, with $\dot V = -\boldsymbol{x}^\top \boldsymbol{Q}^* \boldsymbol{x} < 0$, confirming global asymptotic stability of the closed-loop linear system.

10.2 Open-Loop Instability

The open-loop eigenvalues of $\boldsymbol{A}$ include right-half-plane (RHP) values — the system has at least one positive real eigenvalue per unstable link. LQR moves all closed-loop eigenvalues of $\boldsymbol{A} - \boldsymbol{B}\boldsymbol{K}$ to the left half-plane.

11 MATLAB Implementation Reference

11.1 Script Execution Order

Step	Script	Function
1	`parameters.m`	Define physical constants
2	`linearize.m`	Compute $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$; check rank
3	`lqr_design.m`	Solve ARE, compute $\boldsymbol{K}$; saves `lqr_data.mat`
4	`open_loop_sim.m`	Simulate unstable open-loop
5	`pid_design.m`	Manual PID tuning + simulation
6	`ukf_design.m`	UKF state estimation with noisy measurements
7	`compare_controllers.m`	LQR vs PID performance metrics
8	`animate_pendulum.m`	Generate GIF animations

11.2 Key MATLAB Commands

Command	Purpose
`[A,B] = linearize(params)`	Numerical Jacobian at upright eq.
`rank(ctrb(A,B))`	Verify controllability (must be 8)
`[K,P,e] = lqr(A,B,Q,R)`	Solve ARE, return gain and eigenvalues
`eig(A - B*K)`	Verify closed-loop stability
`ode45(@(t,x)odefun,tspan,x0)`	Nonlinear closed-loop simulation
`expm(A*Ts)`	Discrete-time $\boldsymbol{A}_d$ for MPC

12 Glossary

ARE Algebraic Riccati Equation — governs the optimal LQR cost matrix $\boldsymbol{P}$.

Controllability Ability to drive any state to any other state in finite time.

EKF Extended Kalman Filter — nonlinear estimator using first-order Jacobian linearisation.

LQG Linear Quadratic Gaussian — optimal output-feedback controller (LQR + Kalman filter).

LQR Linear Quadratic Regulator — state-feedback controller minimising a quadratic cost.

MPC Model Predictive Control — receding-horizon constrained optimisation.

Observability Ability to reconstruct all states from output measurements alone.

PID Proportional-Integral-Derivative controller — classical single-loop regulator.

UKF Unscented Kalman Filter — nonlinear estimator using deterministic sigma points.

$\boldsymbol{Q}, \boldsymbol{R}$ LQR weighting matrices — trading state regulation against control effort.

$\boldsymbol{Q}_w, \boldsymbol{R}_v$ Kalman filter noise covariances — process and measurement noise.